Toward a four-dimensional Yang–Mills construction

Notebook 24: inherited interactions and the next-step norm test

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Abstract

We separate an inherited analytic quantum correction into its stationary-background value and its fluctuation increment in the actual ordinary pivot chart. Directional Cauchy bounds the increment by a factor of order gR divided by the available analytic margin, without a carrier-size derivative loss. The exact pivot incidence confines its projected support to the original blocks and their relevant endpoint neighbors. Connected-carrier compression then yields an explicit sufficient inequality for the stronger interaction norm required by the next Gaussian integration. Under that inequality and the existing covariance hypotheses, the inherited contribution to the connected logarithm has the same small factor. The inequality is not yet verified for the actual output decay constants as the block side varies. This result makes that parameter test, and the independent domain-stability problem, explicit.

Notebooks 21–23 provide the classical-reference integral, stationary-background transport, and local gauge covariance. The following argument treats the additional interactions produced by earlier integrations. It keeps their full background term outside the integral and controls the remaining fluctuation effect under a displayed norm budget.

1 A conditional bridge for inherited quantum interactions

This section is a finite-step result with an explicit unverified weight inequality. It uses the actual source fluctuation chart and does not claim that the available constants meet that inequality, that field domains are stable under iteration, or that an RG induction has closed. Fix odd $L \geq 5$, color and compatible periodic source geometry; all displayed constants are uniform in old refinement M and volume, but may depend on L.

1.1 The exact split before the next integration

After stationary transport, the activities are functions of the next-current unit-spacing potential, now denoted a. Their connected carriers H are unit current cells. The following integration groups these cells into L-block Gaussian cells, with dimensions

$$D_G = d(L^d - 1)\dim \mathfrak{g}, \qquad D_{\mathrm{bg}} = dL^d \dim \mathfrak{g}. \tag{1}$$

The first counts retained Gaussian coordinates; the second counts all background link coordinates in a block. The inherited unit-cell activity has at most $D_0 = d \dim \mathfrak{g}$ background scalar coordinates per unit cell.

Write $V_*(W) = e^{ia_*(w)}$, $W = e^{iw}$, for the actual matched stationary branch. The exact source insertion is

$$U = e^{i\Phi_{V_*(W)}(g\xi)} V_*(W). \tag{2}$$

Keep its current background independent initially, and define

$$F(a,z)=\frac{1}{i}\log\left(e^{i\Phi_{e^{ia}}(z)}e^{ia}\right), \qquad F(a,0)=a, \tag{3}$$

$$\Delta E_H(a,\xi)=E_H(F(a,g\xi)|_H)-E_H(a|_H). \tag{4}$$

Then, on the common small domain,

$$\sum_H E_H\left(\frac{\log U}{i}\Big|_H\right)=\sum_H E_H(a_*(w)|_H)+\sum_H \Delta E_H(a_*(w),\xi). \tag{5}$$

The first sum is outside the fluctuation integral. It is not discarded as a vacuum constant unless it is independent of w. There is no factor g^{-2} in (4): E_H is already a dimensionless quantum exponent. Its linear and quadratic fluctuation terms may remain bounded interactions.

1.2 A value estimate without a carrier-size derivative loss

Suppose E_H is bounded holomorphic on the scalar-coordinate product ball $|a|<b$, with $e_H=\|E_H\|_b$, and

$$\sup_i \sum_{H\ni i} e^{a_0|H|}e_H\le\epsilon, \qquad a_0>0, \qquad H \text{ connected}. \tag{6}$$

The audited fixed-L source chart supplies $b_{\rm ch}, z_{\rm ch}, C_{\rm ch}>0$ such that

$$\|F(a,z)-a\|_\infty\le C_{\rm ch}\|z\|_\infty, \quad \|a\|_\infty<b_{\rm ch}, \quad \|z\|_\infty<z_{\rm ch}. \tag{7}$$

All fixed coordinate norm conversions are included in these constants. Choose $b_-<\min(b,b_{\rm ch})$ and set

$$m=b-b_->0, \quad |g|R<z_{\rm ch}, \quad \delta=C_{\rm ch}|g|R<m, \quad u=\frac{\delta}{m-\delta}. \tag{8}$$

Lemma 1.1 (Uniform inherited increment)**.** *On the independent background ball $|a|<b_-$ and retained window $\|\xi\|_\infty\le R$,*

$$\|\Delta E_H\|\le ue_H. \tag{9}$$

The increment is holomorphic in the independent background. If $\delta\le m/2$, then $u\le 2C_{\rm ch}|g|R/m$.

Proof. For fixed allowed (a,ξ), put $h=F(a,g\xi)|_H-a|_H$. The one-variable function $f(\zeta)=E_H(a|_H+\zeta h)$ is bounded by e_H on $|\zeta|<m/\delta$. Its Taylor series gives

$$|f(1)-f(0)|\le e_H\sum_{n\ge1}(\delta/m)^n=ue_H.$$

The actual chart is holomorphic in a, proving the final statement. □

This estimates one directional value increment, not the absolute sum of its multivariate Taylor coefficients. It therefore has no $D_0|H|$ factor and does not replace the coordinate-count argument needed for strict support substitution. No cutoff indicator is differentiated.

1.3 Exact pivot geometry after Gaussian-cell projection

Use positive current bonds, assigned to their initial site, and let π_L be centered L-cube projection. A nonpivot output of the actual map satisfies

$$\Phi_V(z)(b) = z(b), \qquad F(a,z)(b) = \frac{1}{i}\log(e^{iz(b)}e^{ia(b)}),$$

so it reads only that bond. A pivot $p(c)$ for $c = (y, y + Le_\mu)$ lies on the common interface of the two endpoint blocks, with initial site in the y block. Its exact implicit equation reads only the retained links and background links in these two blocks. No other pivot enters the equation, so solving it does not propagate dependence farther.

Define

$$Y(H) = \pi_L(H) \cup \{\text{other endpoint block of } c : p(c) \text{ starts in } H\}. \tag{10}$$

The increment reads only the independent (a,ξ) fibers in $Y(H)$. This carrier is connected, and, with $\chi = d+1$ and $N_{\text{near}} = 2d+1$,

$$|Y(H)| \le \chi|\pi_L(H)|, \qquad \ell(Y(H)) = |Y(H)| - 1, \tag{11}$$

$$\mathbf{1}_{\{B\in Y(H)\}} \le \sum_{i\in H} \mathbf{1}_{\{d_{\text{block}}(B,\pi_L(i))\le 1\}}. \tag{12}$$

Indeed every extra block attaches to $\pi_L(H)$, and at most d outward endpoint blocks are added per projected block. The conservative neighbor count N_{near} is enough for anchoring. A convention reading both incoming and outgoing endpoints may instead use $\chi = 2d+1$. The positive oriented source convention gives the displayed $d+1$.

This uses the actual one-pivot incidence. Enlarging each input cell by an entire radius-$O(L)$ current ball and paying its $O(L^d)$ volume would lose the sharper estimate.

1.4 The explicit sufficient interaction budget

Group equal carriers into the exact exponent

$$Q_Y(a,\xi) = \sum_{H:Y(H)=Y} \Delta E_H(a,\xi). \tag{13}$$

The absolute anchored estimates below justify this grouping. The strict-support, tunable real-baseline theorem of Notebook18 requires the multiplier norm

$$\epsilon_{\text{int}} = \sup_B \sum_{Y\ni B} v_Y \mathcal{K}_\theta^{|Y|} e^{4|Y|+\kappa\ell(Y)}, \qquad v_Y = \sup_{\text{cutoff}} |e^{Q_Y} - 1|, \tag{14}$$

$$\mathcal{K}_\theta = \left(\frac{1+\theta}{(1-\theta)^3}\right)^{D_G/2}, \qquad 0 < \theta \le 1/4. \tag{15}$$

The cutoff-local multiplier is $\chi_Y(e^{Q_Y} - 1)$, extended by zero to all real fluctuations. All additional source restrictions remain present.

Put

$$B_{\text{req}} = 4 + \log\mathcal{K}_\theta + \kappa, \qquad k_L = 1 + \left\lfloor \frac{(L-1)/2 - 1}{2} \right\rfloor. \tag{16}$$

Connected-carrier compression and (11) give

$$|Y(H)| \le \chi 2^d + (\chi 2^d/k_L)|H|. \tag{17}$$

Proposition 1.2 (Conditional inherited interaction re-entry)**.** *Assume the exact source chart,* (6)*,* (8)*, and the numerical inequality*

$$\frac{\chi 2^d}{k_L} B_{\mathrm{req}} \le a_0. \tag{18}$$

Then, with $V_{\max} = uN_{\mathrm{near}}L^d\epsilon$*,*

$$\sup_B \sum_{Y \ni B} e^{B_{\mathrm{req}}|Y|} \|Q_Y\| \le u e^{B_{\mathrm{req}}\chi 2^d} N_{\mathrm{near}} L^d \epsilon, \tag{19}$$

$$\epsilon_{\mathrm{int}} \le e^{V_{\max}} u e^{B_{\mathrm{req}}\chi 2^d} N_{\mathrm{near}} L^d \epsilon. \tag{20}$$

In particular, for $V_{\max} \le \log 2$ *the first prefactor in* (20) *is at most two.*

Proof. Equations (9) and (12) first give

$$\sup_B \sum_{Y \ni B} \|Q_Y\| \le uN_{\mathrm{near}}L^d\epsilon = V_{\max}.$$

Every individual exponent is therefore bounded by $V_{\max}$, and the multiplier estimate is exactly $v_Y \le e^{\|Q_Y\|}\|Q_Y\| \le e^{V_{\max}}\|Q_Y\|$. One cannot retain the smaller individual-label bound ue_H after grouping many H into one Y.

For the weighted bound, (17) and (18) imply

$$e^{B_{\mathrm{req}}|Y(H)|} \le e^{B_{\mathrm{req}}\chi 2^d} e^{a_0|H|}.$$

Use (12) and anchor at the at most $N_{\mathrm{near}}L^d$ current sites in neighboring blocks. The input norm (6) proves (19). Connectedness gives $\ell(Y) = |Y| - 1$, so the target weight in (14) is at most $e^{B_{\mathrm{req}}|Y|}$. The multiplier estimate proves (20). □

Thus the edge-geometry weight is included, not only the $e^{4|Y|}$ term. If another estimate needs a factor $|H|^q$, require a strict margin $\rho = a_0 - (\chi 2^d/k_L)B_{\mathrm{req}} > 0$ and use $\sup_{n\ge 1} n^q e^{-\rho n}$. No extra reserve is assumed without such a budget.

1.5 Comparison of the connected logarithm coefficients

Assume the remaining covariance and precision hypotheses of Notebook18 hold on the independent background ball. Write its common smallness threshold as η_* and its canonical strict connected coefficient estimate as $\|\mathcal{U}\|_{\mathrm{out}} \le C_{\mathrm{out}}\eta$ for $\eta \le \eta_*$. Here the output norm is the theorem's connected support norm in the original independent background variables, not merely the absolute scalar logarithm.

Corollary 1.3 (Inherited contribution to the logarithm)**.** *Let* η_{base} *include the Gaussian tail and the classical, measure and gauge interaction norms. If*

$$\eta_{\mathrm{base}} \le \eta_*/4, \qquad \epsilon_{\mathrm{int}} \le \eta_*/4,$$

then the canonical coefficients with and without the inherited multipliers satisfy

$$\|\mathcal{U}_{\mathrm{full}} - \mathcal{U}_{\mathrm{base}}\|_{\mathrm{out}} \le 4C_{\mathrm{out}}\epsilon_{\mathrm{int}}. \tag{21}$$

Proof. If $\epsilon_{\mathrm{int}} = 0$ the statement is immediate. Otherwise multiply only each inherited bounded multiplier $\chi_Y(e^{Q_Y} - 1)$ by a complex parameter λ, keeping the base family fixed. At $\lambda = 0$ and 1 this gives the two exact integrals. It does not replace Q_Y by λQ_Y. Base and inherited factors with the same carrier are retained as distinct labelled hyperedges, as permitted by the overlap theorem; no

cancellation or division by a base factor is used. For $|\lambda| \le R_\lambda = \eta_*/(2\epsilon_{\rm int}) \ge 2$, the total budget is at most $3\eta_*/4$. The canonical coefficient vector $\mathcal{U}(\lambda)$ is holomorphic in the output Banach space and bounded by $C_{\rm out}\eta_*$. Summing its Cauchy coefficient bounds gives

$$\|\mathcal{U}(1) - \mathcal{U}(0)\|_{\rm out} \le \frac{C_{\rm out}\eta_*}{R_\lambda - 1} \le 4C_{\rm out}\epsilon_{\rm int}.$$

□

The fluctuation-independent sum $\sum_H E_H(a)$ remains outside this comparison. It and the new connected coefficients are evaluated at $a_*(w)$ afterwards and transported on a common smaller domain by the transport theorem of Notebook 22.

1.6 The open parameter and domain tests

The tunable density estimate supplies

$$\log \mathcal{K}_\theta \le \tfrac{5}{2} D_G \theta, \qquad \theta = \frac{1}{4D_G} \Longrightarrow\ B_{\rm req} \le \frac{37}{8} + \kappa. \tag{22}$$

Without this tuning, a cost $\log K_G = O(D_G)$ in the sufficient interaction norm would contribute $O(L^{d-1})$ to (18). Tuning removes that particular artificial dimension loss. It does not remove the smaller covariance-edge radius, the required $O(\theta)$ precision neighborhood, or the $2^{D_G/2}$ Gaussian tail prefactor. The actual independent background ball must meet those strengthened precision conditions. For the actual linkwise color-ball cutoff $\chi_B = \prod_{b\in B} \mathbf{1}_{\{|\xi_b| \le R\}}$, the bad event is contained in $\{\|\xi_B\|_2 > R\}$. The valid Euclidean-block tail therefore uses threshold R. The good event only implies $\|\xi_B\|_2 \le \sqrt{d(L^d-1)}R$; this opposite-direction bound cannot improve the bad-event threshold. The chart uses link supremum and uses the original R directly. No Euclidean block window is silently identified with the source linkwise window.

For each fixed L, a logarithmic cutoff $R(g)$ can grow while $gR(g) \to 0$, so the chart margin and a fixed Gaussian tail threshold can both be met by sufficiently small g. This does not prove the required all-scale coupling recursion.

More fundamentally, the preceding integration supplies an actual reserve $a_0 = a_0(L)$. Existing fixed-L results give positivity, but no quantitative large-L lower bound proving

$$k_L a_0(L) \ge \chi 2^d [4 + \log \mathcal{K}_\theta(L) + \kappa(L)]. \tag{23}$$

Positivity alone is compatible with $a_0(L) = e^{-L}$ and cannot establish this test. This is a missing estimate for the actual available constants, not a counterexample to Yang–Mills theory. A quantitative locality estimate as L varies, or an integration theorem with a smaller fixed carrier cost, is a concrete next target.

The domain is also not automatic. One needs $\|a_*(w)\|_\infty \le C_a(L)r \le b_-$, and coefficient transport may require $C_a(L)r/b$ of order $a_0/D_{\rm bg}$. No L^{-1} contraction in link-angle supremum has been proved; a pure-gauge stationary branch can have crossing-link angle of order w. Repeated use of the one-step radius losses is therefore not a stable domain construction. The constants in (20) need not be contractive, either.

Finally, strict locality must be applied in the correct order. On independent a, (13) has the strict joint (a, ξ) carrier Y. Apply the Gaussian theorem on that product domain, then substitute the actual $a_*(w)$ into its holomorphic identity. Substitution first generally makes the background dependence nonlocal in w; its rooted connecting trees need their own support budget, and Y alone is no longer a strict w-support. A term independent of w but dependent on ξ is still an interaction, not a vacuum constant. Neither order removes the later radius and weight losses.

The proposition and corollary establish a usable conditional inherited-interaction module. The re-entry test, stable domains, relevant and marginal extraction, running couplings, and large-field and observable sectors remain separate obligations.

In the extended construction, Notebook 26 supplies the support-weight test using independent fixed auxiliary cells. It re-expands the classical precision and Gaussian normalization on those cells and recovers a common positive quantum carrier exponent. That argument resolves this particular norm-weight interface; the domain and amplitude recurrence remains open.

References

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